

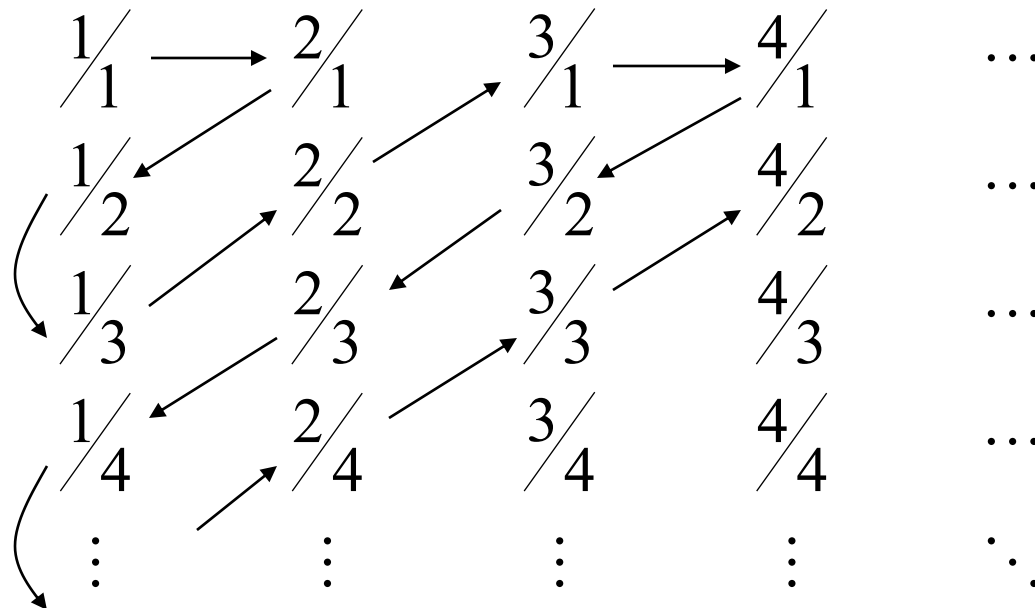
Mathematical Induction

Cardinality & Counting

- $|A| \leq |B|$ if there is an injective function from A to B
- A set A is said to be **countable** if there is an injective function from A to $\mathbb{N}$
 - E.g. $f(x) = \begin{cases} -2x & \text{if } x < 0 \\ 2x + 1 & \text{otherwise} \end{cases}$ is an injective function from $\mathbb{Z}$ to $\mathbb{N}$
- Otherwise, it is **uncountable**

More Countable Sets

- *The set of positive rational numbers is countable*
- *Every rational number can be represented as a fraction $\frac{p}{q}$*
We do not insist that p and q do not have a common divisor



- *This gives an injection from $\mathbb{Q}^+$ to $\mathbb{N}$. The converse injection is $f(x) = x + 1$*

Uncountable Sets

● To show something is uncountable, need to show there *does not exist* an injective function from A to $\mathbb{N}$

● Usually done by *contradiction*

● Are the real numbers countable? Assume yes!

1. $a_{10}.a_{11}a_{12}a_{13}a_{14}a_{15}a_{16}a_{17}\dots$
2. $a_{20}.a_{21}a_{22}a_{23}a_{24}a_{25}a_{26}a_{27}\dots$
3. $a_{30}.a_{31}a_{32}a_{33}a_{34}a_{35}a_{36}a_{37}\dots$
4. $a_{40}.a_{41}a_{42}a_{43}a_{44}a_{45}a_{46}a_{47}\dots$
5. $a_{50}.a_{51}a_{52}a_{53}a_{54}a_{55}a_{56}a_{57}\dots$
- $\vdots$

Here the a_{ij} are digits $0, 1, 2, \dots, 9$

Let

$$b_i = \begin{cases} 4, & \text{if } a_{ii} \neq 4, \\ 5 & \text{otherwise} \end{cases}$$

● Is the number $0.b_1b_2b_3b_4b_5b_6b_7\dots$ in this list?

Cantor's Theorem

● **Theorem (Cantor).** For any set $|P(A)| > |A|$.

Proof.

It is easy to see that $|P(A)| \geq |A|$. (Find an injection $A \rightarrow P(A)$)

Prove $|P(A)| \neq |A|$. Suppose there is a bijection $f: A \rightarrow P(A)$.

Consider the set $T = \{a \in A \mid a \notin f(a)\}$. We will show that T is not in the range of f

If T is in the range of f , then there is $t \in A$ such that $f(t) = T$.

Either $t \in T$ or $t \notin T$.

If $t \in T$ then $t \in f(t)$, and we get $t \notin T$.

If $t \notin T$ then $t \in T$.

Q.E.D.

Familiar?

Cantor's Theorem (cntd)

- *This method is called Cantor's diagonalization method*
- *The cardinality of $P(A)$ is denoted by $2^{|A|}$*
- *Thus, we obtain an infinite series of infinite cardinals*

$$|\mathbb{N}| = \aleph_0$$

$$2^{\aleph_0} = \aleph_1 \quad (= |\mathbb{R}|)$$

$$2^{\aleph_1} = \aleph_2$$

$$\vdots$$

Continuum Hypothesis

- *We just proved that $\aleph_0 < |\mathbb{R}|$. Does there exist a set A such that $\aleph_0 < |A| < |\mathbb{R}|$?*
- *The negative answer to this question is known as the **continuum hypothesis**.*
- *Continuum hypothesis is the first problem in the list of Hilbert's problems*
- *Paul Cohen resolved the question in 1963.
...It can't be proven or disproven in ZFC.*

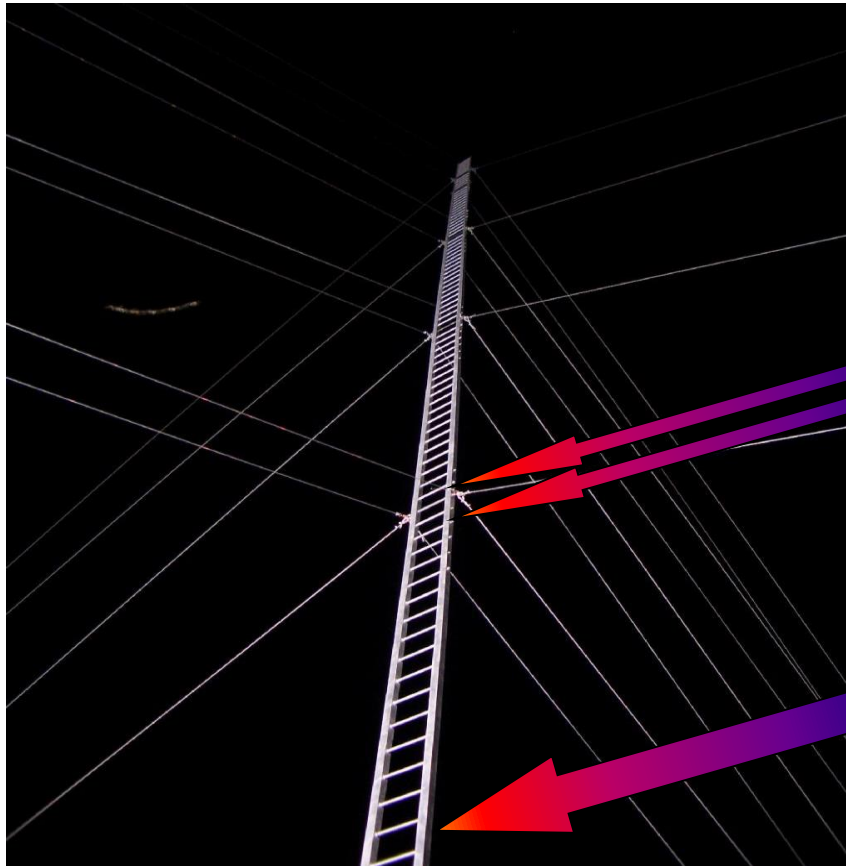


The Smallest Infinite Set

- *What about an (infinite) set smaller than $\aleph_0$?*
- *Theorem.*
If A is an infinite set, then $|A| \geq \aleph_0$
- *Proof requires mathematical induction...*

Principle of Mathematical Induction

- Idea is to prove properties about infinite *but countable* sets by showing that each element *implies the next*



Climbing an infinite ladder

We can reach every step of it, if

For all k , standing on the rung k we can step on the rung $k + 1$

We can reach the first rung

Principle of Mathematical Induction

- *Principle of mathematical induction:*

$$(P(1) \wedge \forall k (P(k) \rightarrow P(k + 1))) \rightarrow \forall n \in \mathbb{Z}^+ P(n)$$

- *Proof by induction* uses the principle of induction to prove that a property $P(n)$ holds for every positive integer n

- Comprises two steps:

Basis step: We show that $P(1)$ is true.

Inductive step: We show that if $P(k)$ is true for some positive integer k , then $P(k+1)$ is also true

- Explicitly, for the inductive step, assume $P(k)$ is true for some k (the *inductive hypothesis*) and show that $P(k+1)$ is also true

The Domino Effect



- *Goal: Show that all dominos fall if we push the first:*
- *Basis Step: The first domino falls because we pushed it*
- *Inductive step: Assume the k th domino fell (*inductive hypothesis*). Then the $(k+1)$ th domino will be hit and also fall*

Summation

- *Prove that the sum of the first n natural numbers equals $\frac{n(n+1)}{2}$ that is*

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$
- *$P(n)$: the sum of the first n natural numbers equals $\frac{n(n+1)}{2}$*
- *Basis step: $P(1)$ means*

$$1 = \frac{1(1+1)}{2}$$
- *Inductive step: Assume the inductive hypothesis, $P(k)$ is true, i.e.*

$$1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$$

Prove $P(k+1)$:

$$1 + 2 + 3 + \cdots + k + (k+1) = \frac{(k+1)((k+1)+1)}{2}$$

$$1 + 2 + 3 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) \quad \text{By inductive hyp.}$$

$$= \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$$

More Summation

- Prove that $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$
- Let $P(n)$ be the statement ' $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ ' for the integer n

● Basis step: $P(0)$ is true, as $2^0 = 1 = 2^{0+1} - 1$

● Inductive step: We assume the inductive hypothesis

$$1 + 2 + 2^2 + \cdots + 2^k = 2^{k+1} - 1$$

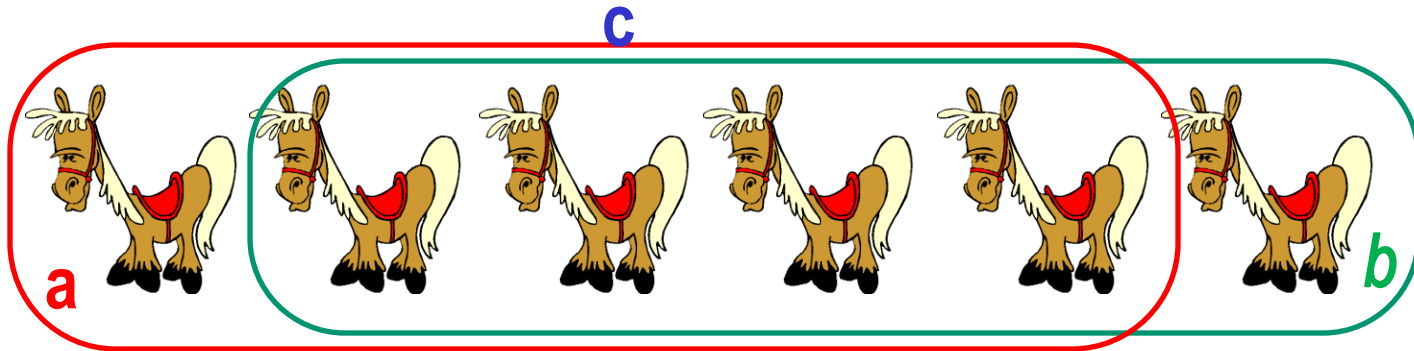
and prove $P(k + 1)$, that is

$$1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} = 2^{(k+1)+1} - 1 = 2^{k+2} - 1$$

$$\begin{aligned} \text{We have } 1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} &= (2^{k+1} - 1) + 2^{k+1} \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{k+2} - 1 \end{aligned}$$

No Horse of Different Color

- Prove that in any herd of n horses all horses are of the same color



- Basis step: $n = 1$, clear
- Inductive step: Inductive hypothesis: Suppose in any herd of k horses all horses have the same color.
- Take a herd of $k + 1$ horses, and pick horses a and b there
 Take two 'subherds' containing k horses each such that one contains a , but not b , and the other contains b , but not a
 The intersection contains some horse c . By inductive hypothesis a has the same color as c and b has the same color as c

The Cardinality of the Power Set

- We have proved that, for any finite set A , it is true that $|P(A)| = 2^{|A|}$
- Let $Q(n)$ denote the statement 'an n -element set has 2^n subsets'
- Basis step: $Q(1)$, then $P(A) = \{\emptyset, A\}$ which has size 2
- Inductive step. We make the inductive hypothesis, a k -element set A has 2^k subsets

We have to prove $Q(k + 1)$, that is if a set A contains $k + 1$ elements, then $|P(A)| = 2^{k+1}$

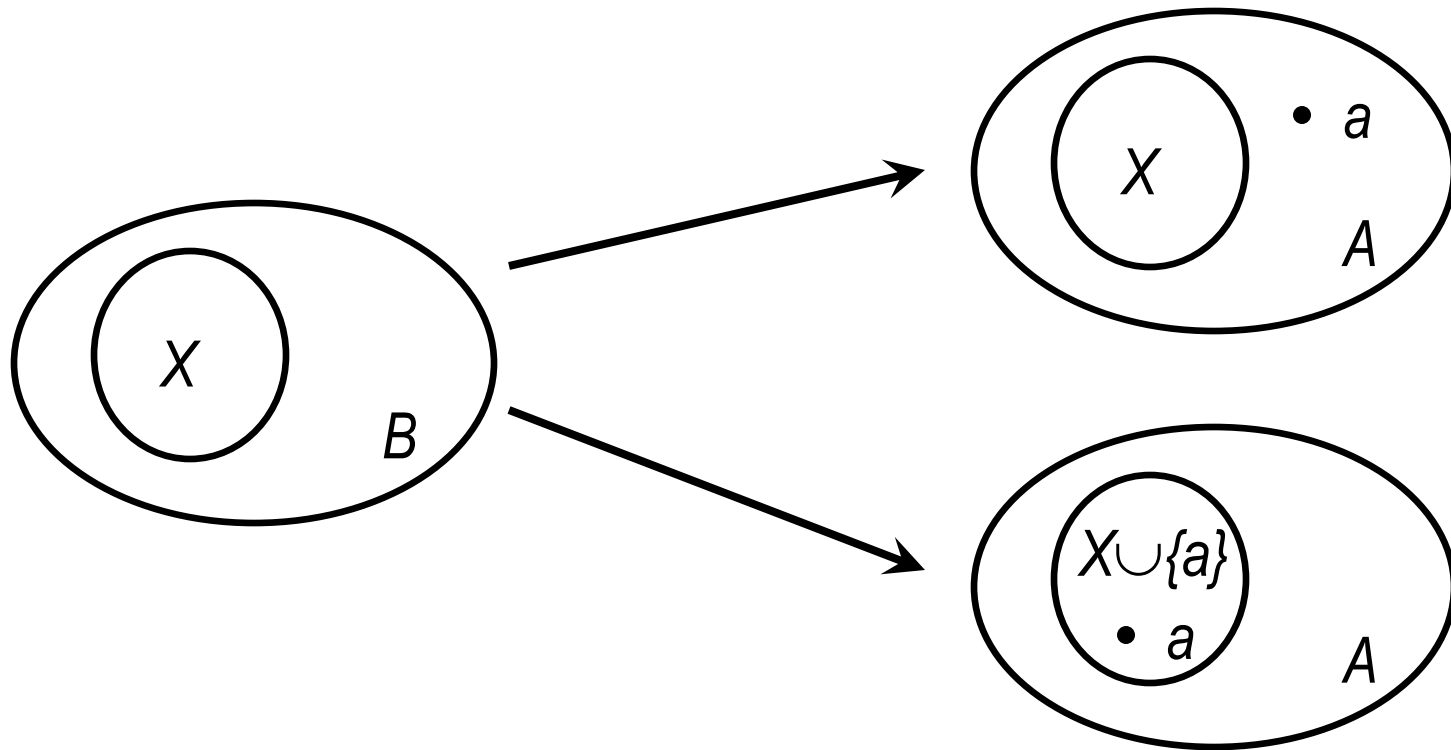
Fix an element $a \in A$, and set $B = A - \{a\}$.

The set B contains k elements, hence $|P(B)| = 2^k$

Every subset X of B corresponds to two subsets of A :

$$X \text{ and } X \cup \{a\}$$

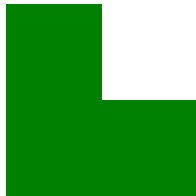
The Cardinality of the Power Set (cntd)



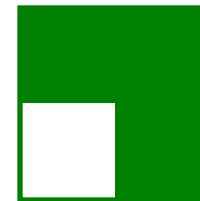
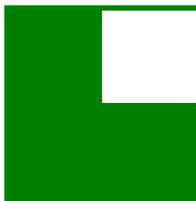
Therefore, $|P(A)| = 2 \cdot |P(B)| = 2 \cdot 2^k = 2^{k+1}$

Triomino

- Let n be a positive integer. Show that every $2^n \times 2^n$ checkerboard with one square removed can be tiled using triominos



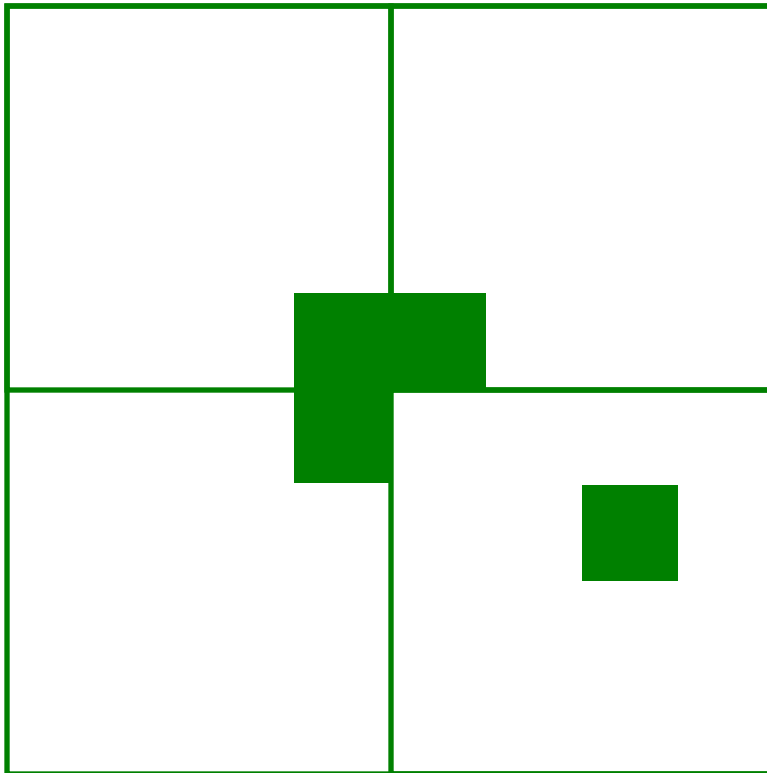
- $P(n)$ denotes the statement above
- Basis step: $P(1)$ is true, as 2×2 checkerboards with one square removed have one of the following shapes



Triomino (cntd)

● *Inductive step: Suppose that $P(k)$ is true that is every $2^k \times 2^k$ checkerboard with one square removed can be tiled with triominos.*

We have to prove $P(k + 1)$, that is, every $2^{k+1} \times 2^{k+1}$ checkerboard without one square can be tiled.



Split the big checkerboard into 4 half-size checkerboards

WLOG assume the bottom right is missing a square. Put one triomino as shown.

We have 4 $2^k \times 2^k$ checkerboards, each without one square. By the induction hypothesis, they can be tiled.

Practice

Exercises from the Book:

7th edition: 5, 14, 19, 31 (page 329 – 330)

8th edition: 5, 14, 19, 31 (page 350 – 351)